

ASHENFALL — Implementation Pipeline

Companion document to DESIGN.md (v0.7) · The full build pipeline, written down so nothing is forgotten Prepared 2026-09-26

This document answers one question: **when we actually build this, in what order do we touch what, and where does every piece of content live?** It is the operational counterpart to PHASES.md (which says *what* each phase adds) — this says *how*.

Five subsystems, in the order they must be built:

- | | |
|-------------------|--|
| 1. Mod management | — the container everything else live |
| 2. Worldgen | — the world must exist before anything |
| 3. Progression | — the rules that gate content |
| 4. Story | — the content the gates protect |
| 5. Optimization | — the budget that all of the above m |

They are listed in build order, but optimization runs *throughout* — it is not a final phase.

Part 0 — The workspace layout

0.1 Repository structure

```
ashfall/
```

```

├─ README.md # how to use the builder
├─ build.bat / build.sh # wrappers around tools/
├─ DESIGN.md PROGRESSION.md LORE.md MODS-INTERACTIONS
├─ RECOMMENDATIONS.md PHASES.md PIPELINE.md DUNGEON-M
├─ tools/
│   └─ build.py # THE generator: sync /
│   └─ modlist.toml # THE source of truth: e
├─ pack/ # ← the pack, as text. C
│   └─ pack.toml # packwiz manifest: na
│   └─ index.toml # packwiz mod index: o
│   └─ mods/ # one .pw.toml per mod
│   └─ pending-mods.toml # the mods that need a
│   └─ CHANGELOG.md
│   └─ overrides/
│   └─ config/ # configs shipped to the
│   └─ data/ # — DATAPACK LAYER —
│       └─ ashfall/ # our namespace: recip
│       └─ minecraft/ # vanilla overrides (l
│       └─ <modid>/ # per-mod overrides
│   └─ kubejs/
│       └─ server_scripts/ # recipes, events, sta
│       └─ startup_scripts/ # items, blocks, tags,
│       └─ client_scripts/ # tooltips, JEI, HUD
│       └─ data/ # KubeJS-generated dat
│   └─ patchouli_books/
│       └─ pilgrims_journal/ # 24 pages
│       └─ lore_codex/ # 24 entries
│   └─ ftbquests/ # chapters 0-8, quest
│   └─ resourcepacks/
│       └─ ashfall_loot_beams/ # rarity colours match
│   └─ shaderpacks/ # EMPTY BY DESIGN - th
├─ build/ # ← SCRATCH. Excluded fr
│   └─ mods/ # the JARS (~400-600 M
│   └─ export/ # the distributable .m
│   └─ cache/ # Modrinth API respons
└─ profiles/
    └─ performance.toml
    └─ compat.toml # Acedium off, stock E
    └─ magic.toml # Monsters & Spellbook

```

Why packwiz and not a hand-assembled zip: packwiz keeps every mod as a separate line in `index.toml` with a pinned hash and URL. That means (a) a git diff shows exactly which mod changed, (b) updating one mod is a one-line change, (c) the export to a CurseForge/Prism `.zip/.mrpack` is a single command, and (d) a crash's stack trace can be traced to a hash immediately.

Why generated and not hand-edited: 132 mods is past the point where a human maintains `index.toml` by hand. `tools/build.py` reads `tools/modlist.toml` and writes the whole packwiz project — see §1.5. Editing `index.toml` directly is a bug, not a shortcut.

0.2 The commit rule

Every change to this repository gets one commit with a message in the form:

```
[M3c] add The Graveyard + Lich's Bargain rework
      - loot: bone staff fragments injected into Ruins ch
      - kubejs: villager-blood ritual replaced with lich-
      - gate: M3c #4 (magic chain) still failing - Inscri
```

The milestone tag in brackets is mandatory. It is how we reconstruct *when* a regression entered.

0.3 The three golden rules

1. **One mod per commit.** Two mods in one commit means two suspects when it breaks.
 2. **Never edit a config without saying why** in the commit message. Configs are 60% of the pack's personality.
 3. **Never add a worldgen mod to an existing world.** Worldgen changes require a new world, always.
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Part 1 — Mod management

1.1 Version pinning policy

Class of mod	Pin	Why
Unofficial ports (Acedium Sodiumized, Alex's Caves, Enigmatic Legacy+, Millénaire forks)	Exact hash. Never auto-update	These are community builds with no upstream support; an "update" can be a different mod entirely
Libraries (Lionfish API, Curios, Sophisticated Core, Player Animator, Terrablender)	Exact hash. Never auto-update	Every content mod depends on one specific version
Active development (Monsters & Spellbooks, any mod with 0.0.x versions)	Exact hash , with a monthly re-check	The author has stated content will be reworked
Stable content (Cataclysm, Twilight Forest, Project MMO, YUNG's suite)	Pin the minor , allow patch updates inside a milestone boundary only	Patch updates are usually fixes; minor updates change recipes/loot
Performance (Embeddium + extras, ModernFix, FerriteCore, etc.)	Pin exactly , then re-test as a batch once per milestone	These interact with each other in non-obvious ways

`packwiz refresh` is **only** run at a phase boundary, never mid-phase, and its diff is reviewed before commit.

1.2 The update protocol

When a mod has a new version:

1. **Read the changelog.** If it mentions recipes, loot tables, structures, config format, or a dependency bump → treat as a breaking change.
2. **Check the port-risk register** (MODS-INTERACTIONS §7). If the mod is on it, stop — this needs a dedicated session.
3. **Update one mod**, boot, and run the phase's gate tests.
4. **If anything fails, revert the single line in `index.toml`** and record it in CHANGELOG.md as "held at X.Y.Z — reason".
5. **Never update two mods at once** unless both are pure performance mods.

1.3 Mod budget enforcement

The budget is **≤130 content mods + ≤18 utility/performance mods**.
Enforcement:

- `index.toml` is the source of truth. A pre-commit check counts entries.
- Adding a mod requires **removing a mod or naming the cut**. The cut order (DESIGN §6.11 and §13) is the named list of what goes first: Currents of Trade → Sable/Sablewaves → Streams Reflowing → Supplementaries → Comforts → Tool Belt → Farmer's Delight → Boatload → Flowing Fluids → Monsters & Spellbooks → SWEM (*Monsters & Spellbooks sits near the end of that list — it's the newest risk, and dropping it never breaks the base mod.*)
- Performance mods are capped at 18 because past that the interaction cost exceeds the gain.

1.4 Config management

- All configs live in `config/` **and** `overrides/config/`. `config/` is the working copy; `overrides/config/` is what ships.
- Any config change that affects **worldgen, water, structure spawn rates, or entity caps** is tagged `[LOCK]` in the commit message, because those require a new world or a re-test.
- Three profile variants exist as config/option sets, not mod changes:
 - `performance` — Acedium on, all culling on, no Distant Horizons, entity caps tight
 - `compat` — Acedium off, stock Embeddium, for GPUs without mesh shader support
 - `magic` — Monsters & Spellbooks disabled, for stability testing

1.5 The pack builder

The pack is **generated, never hand-assembled**. `tools/build.py` (stdlib Python 3.11+, no dependencies) turns one manifest into a packwiz project.

```
tools/modlist.toml    THE source of truth: every mod, its
    |
    v
python tools/build.py sync        resolves versions + s
    |
    +--> pack/pack.toml           packwiz manifest, MC
    +--> pack/index.toml         one line per mod: url
    +--> pack/mods/*.pw.toml     per-mod packwiz metad
    +--> pack/pending-mods.toml  whatever could not be
    |
    v
python tools/build.py fetch        downloads into build/
    |
    v
python tools/build.py export      -> build/export/ashfa
```

Commands: `list · sync · fetch · verify · count · export`. All

accept `--phase M0` to restrict to one build phase, so a regression can always be traced to the phase that introduced it.

The space split (this is the answer to "how do we build without filling the workspace"):

Path	Contents	Size	In the snapshot?
pack/	pack.toml, index.toml, mods/*.pw.toml, config/, kubejs/, overrides/	~1 MB	Yes — this is the pack
build/ mods/	the JARs	~400–600 MB	No — build/ is excluded by name
build/ export/	the .mrpack	~450 MB	No
build/ cache/	Modrinth API responses (15-min TTL)	~1 MB	No

So the durable workspace holds **text**, and the heavy artifacts are disposable and re-fetchable at any time with one command.

Mod resolution status: 131 of 132 mods resolve automatically — 118 from Modrinth (slug → newest 1.21.1 NeoForge/Forge build → sha512), 12 from CurseForge (file id → `mediafilez.forgecdn.net` URL), and 1 from a GitHub release. The only one left is **GhostSwap**, our custom mod, which is a `direct` entry with a TBD URL — the **source is now written** in `ghostswap/` (see `ghostswap/README.md`), so M1 is a `./gradlew build` plus a jar rename rather than an authoring task.

CurseForge is reachable after all. The HTML files pages are Cloudflare-gated, but the site's own JSON endpoint `www.curseforge.com/api/v1/mods/{projectId}/files` is open, and project IDs are recoverable from search results. Two gotchas that cost real time: the CurseForge slug is often *not* the Modrinth one (`ftb-quests` → `ftb-quests-forge`, project 289412), and the JAR filename is a URL path segment, so `+` must be percent-encoded as `%2B` or the CDN returns 403.

Windows and POSIX wrappers: `build.bat` and `build.sh` at the workspace root. Full instructions in `README.md`.

1.6 The boot-test loop

Every phase's work is verified with this exact loop, in order:

1. **Cold boot** (fresh JVM launch) → check the log for warnings and mixin errors.
2. **World load** → existing test world, no new world unless the phase is `worldgen`.
3. **Phase gate** (`PHASES.md`) → run every numbered test.
4. **Log scan** → grep the log for `ERROR`, `Failed to`, `Mixin apply failed`, and any mod id in the port-risk register.
5. **Commit** with the milestone tag and the gate result.

Crash triage order: read the first `Caused by`: in the stack trace → identify the mod id → check whether that mod was added in the last commit → if yes, revert that commit before anything else.

Part 2 — Worldgen

Worldgen is the one subsystem with a hard ordering constraint: **the water stack must be installed before the world exists**.

2.1 The order (never change it)

1. Perf stack only (M0)
2. Water stack: `Dynamic Waters` + `Flowing Fluids` + `Boatload`
3. `Terralith` + `Terrablender` (+ `Tectonic off`)
4. `Explorify`, `Dungeons and Taverns`, `Towns and Towers`, `Whe`

- 5. YUNG's suite (×9) + Hopo (portals/underwater ruins) +
- 6. The Graveyard, Structory, Dimensional Dungeons
- 7. Iron's Spells structures (Catacombs, Lost Citadel, Ech
- 8. *(Ancient Warfare 2 is out – no 1.21.1 build. Its prov
- 9. Generate the world. Never add or remove a worldgen mod
- 10. Chunky starter pregen ~10k × 10k, then on-demand preg

Why this order: Dynamic Waters rewrites river generation and must be the first worldgen mod in. Every structure mod after it places into a world whose rivers already exist, so we can check that structures don't spawn inside reworked rivers.

2.2 Density tuning loop

For each structure mod, run this and record the numbers:

Metric	Target
Structures per 1,000 × 1,000 blocks	15–30 (enough to find one every few minutes)
Vanilla structure types fully replaced	15/15 (Tier A)
Duplicate mineshafts	0 (YUNG's only)
Repurposed Structures variant rate	Tuned so a displacement scar is <i>rare enough to be a discovery</i>
Explorer's Compass blacklist entries	Every noise structure excluded

Tuning method: fly a 5,000-block survey line in each of four biomes, count structures, adjust the mod's spawn-rate config, regenerate a fresh test region (never the shipping world), repeat.

2.3 The four discovery systems and how

they're built

System	Implementation	Data source
Explorer's Compass	Config blacklist of noise structures; the compass then only points at Landmark-chain structures	<code>config/explorerscompass/</code> + a datapack structure tag
Cartographer's Maps	Custom items injected into dungeon loot tables, each holding a structure coordinate	<code>overrides/data/ashfall/loot_tables/</code> + KubeJS item with structure NBT
Rumours	FTB Quests chapters + KubeJS dialogue that names a landmark without a marker	<code>ftbquests/</code> + <code>kubejs/server_scripts/rumours.js</code>
The Pilgrim's Journal	Patchouli pages that double as treasure maps	<code>overrides/patchouli_books/pilgrims_journal/</code>

Never soft-locked: every landmark must be reachable by ≥ 2 of these four. The rule is checked in M7's gate and re-checked in M9.

2.4 Distance bands and scaling

Band	Distance	Mob scaling	Threat tiers available
Heartland	0–1,500	$\times 1.0$	I–III
The Reaches	1,500–6,000	$\times 1.3\text{--}1.8$	I–V

The Frontier	6,000–20,000	×2.0–3.0	I–VII
The Ashen Wastes	20,000+	×3.5+	I–VII

The world has **no border and no size limit** — the bands exist so that "further out" is a decision, not a surprise. Nothing story-critical lives past 20,000 blocks.

2.5 Pregen policy

- **Starter region:** Chunky, ~10k × 10k, at build time only.
- **Beyond that:** on-demand pregen — the pack ships a Chunky config that pregens around the player as they travel, with a cap so it never fills the disk.
- **Never** whole-world pregen on an infinite world. It has no end and it isn't a feature.

Part 3 — Progression

Progression is the layer that decides *when* content is allowed to exist. It is built from five interlocking pieces, and the order matters because each one feeds the next.

3.1 The five pieces, in build order

1. Skill tracks	(Project MMO)	→ what the player
2. Tier ladder	(gear + stations)	→ what materials
3. Recipe gating	(KubeJS)	→ what can be cra

4. Quest flags (FTB Quests) → what has been u
5. Landmark discovery(4 systems, §2.3) → where the playe

3.2 Skill tracks

Eight vanilla-plus tracks: **Mining, Woodcutting, Combat, Farming, Fishing, Smithing, Agility, Magic.**

- Skills level by *use* (Mining from mining, Combat from killing, Magic from casting).
- Each level grants real attributes via Project MMO's config — not just a number.
- **The magic track runs in parallel** and is optional; it opens at Tier 3 (PROGRESSION §4A).
- FallingTree's logs-per-cut is gated by Woodcutting, so the QoL mod and the grinding loop are the same system.

3.3 The tier ladder

Iron/Diamond → Netherite → Witherite → Aether → Arcane →

Each tier's entry is a **discovered landmark**, never a level number. Each tier's crafting station is a milestone reward:

Tier	Station	From
Netherite	vanilla smithing	vanilla
Witherite	Mechanical Fusion Anvil	Ignis
Aether	Aether Altar / Freezer	Aether
Arcane	Scroll Forge → Inscription Table → Arcane Anvil	Iron's Spells
Hollow Court	Hollow Court infusion	Twilight Forest

3.4 Recipe gating (KubeJS)

Every modded recipe is wrapped in a gate. The gate types, in order of preference:

1. **Skill gate** — `smithing >= 20` etc. The main mechanism.
2. **Boss-material gate** — requires a specific drop, so the recipe is a reward for a kill.
3. **Quest-flag gate** — requires an FTB Quests chapter flag.
4. **Standing gate** — requires nation standing $\geq X$ (only for nation gear lines).

Never gate on: luck, RNG, time played, or anything a player cannot see coming. The rule is that a player should always be able to answer "what do I need to do next?"

3.5 The gating audit

Every gate must answer these five questions. If any answer is "no", the gate is wrong:

1. Can the player see what's required?
2. Is the requirement obtainable without an unreachable item?
3. Is there a second route if the first is missed?
4. Does failing the gate leave a usable fallback?
5. Does the gate survive a New Game+ reset?

This audit is run per gate in M8's phase, and re-run in M9's playthrough pass.

Part 4 — Story

The full narrative implementation guide — dialogue engine choice, NPC templates, the rumour register, boss monologue cutscenes, resource packs, and the verdict on custom mods — is [NARRATIVE-SYSTEMS.md](#). This part covers the registers and the delivery chain.

The story is content, and content needs a pipeline like everything else. The direction of travel is always: **LORE.md** → **structure placement** → **item/loot injection** → **Patchouli** → **FTB Quests** → **tuning**.

4.1 The content registers

Every piece of story content is a numbered register entry. These are the pack's actual deliverables:

Register	Count	Lives in	Built in
The Empire's Annals	12 texts	datapack structures + Patchouli	M7
The Pilgrim's Journal	24 pages	Patchouli book	M7
The Lore Codex	24 entries	Patchouli book	M3c + M7
Ember Runes	9 items	KubeJS + structure loot	M3c + M7
Boss lore entries	21	music discs + Patchouli	M6
Nation chronicles	9	LORE.md §5 → FTB Quests	M4
FTB Quests chapters	9 (0–8)	ftbquests/	M7

Rule: no register entry may ship as a placeholder. An empty journal page is worse than a missing one.

Narrative tooling (see NARRATIVE-SYSTEMS.md): dialogue is JSON in `data/ashfall/adm_dialogues/` (Aviel's Dialogue Mod), quest logic that needs flags is KubeJS via FTB XMod Compat, and monologues are KubeJS. **Two new mods total** — everything else is JSON, datapack and art.

4.2 The delivery chain for one piece of lore

Example — Annals entry #4, found in the Mayan Sun-Vault:

1. **LORE.md** — the text is written (the Mayan Deep's account of the fall).
2. **Structure** — the Jungle Temple is reworked by YUNG's Better Jungle Temples and named *The Mayan Sun-Vault* (M3c).
3. **Loot injection** — a datapack loot table adds the Annals page to that structure's chest pool, with a low probability and a "guaranteed once" flag so it can't be permanently missed.
4. **Patchouli** — the page is a page in `pilgrims_journal`, cross-referenced in `lore_codex`.
5. **FTB Quests** — a chapter objective ("find the fourth Annals text") with a reward that points at the next clue.
6. **Gate check** — M7's gate: all 12 obtainable, in order, and all 12 required for the Cathedral of Ash.

Every one of the 24 journal pages and 12 Annals follows this exact six-step chain. That is the whole story pipeline.

4.3 The naming rule

Every structure, item, and NPC that carries story gets an **in-world name** before it gets a code name:

Code name	In-world name
monster dungeon	<i>The Delver's End</i>
stronghold	<i>The Seal Vault</i>
mineshaft	<i>The Deep Delve</i>
jungle temple	<i>The Mayan Sun-Vault</i>
ocean monument	<i>The Drowned Choir</i>
nether fortress	<i>The Burning Bastion</i>
end city	<i>The Emperor's Last City</i>

A player should never say "I found a desert temple". They should say "I found a Seljuk Sand-Vault".

4.4 The standing system

Built in M4 as KubeJS state + FTB Quests hooks:

- Range –100...+100 per nation, nine nations.
- Drivers: quest choices, tribute, raiding vs defending, sparing vs killing champions, and **class** (a Hexer starts at –40 with the Witchbane).
- Cross-faction: helping one nation's enemy costs standing with them.
- **You cannot be champion of all nine.** This is a hard rule, verified in M4's gate, and it is what makes the ending choice real.

4.5 The endings

Four endings, each with an explicit, auditable requirement:

Ending	Requires
--------	----------

The Empire Rekindled	High standing with the Byzantine Choir
The Pilgrim's Road	High standing with ≥ 3 non-Choir nations
The Tenth Ember	All 24 journal pages + one nation at +100 + all 12 Hollow Challenges
The Ashen Throne	All nine Ember Runes → the Hidden Cathedral

The requirements are checked as **counters**, not as narrative state — a counter can be audited, a narrative state cannot.

Part 5 — Optimization

Optimization is not a phase; it is a running budget checked at every gate. The target is **60+ FPS at render distance 10–12, no shaders, on an NVIDIA 3500-series-class GPU, with 6 GB of RAM allocated.**

5.1 The budget

Metric	Target	Hard ceiling
FPS (overworld, render 12)	60+	—
FPS (boss arena, particles)	60+	—
FPS (flooded structure, water physics)	55+	—
RAM	6 GB	8 GB
Content mods	≤ 130	—
Utility/performance mods	≤ 18	—

Tick time (mob farm + water volume)	< 45 ms/t	50 ms/t
Shaders	Never	—

5.2 What makes it fast

Layer	Mods / setting
Rendering	Embeddium + Extra/++ , Acedium Sodiumized (Nvidia, GTX 1600+/RTX 20xx+, auto-disables under shaders, silent fallback to stock Embeddium)
Memory	FerriteCore, ModernFix (dynamic resources + faster item rendering on)
GUI	ImmediatelyFast (<code>experimental_screen_batching</code> on if it passes M0's GUI test)
Entities	Entity Culling, MoreCulling, Clumps, entity caps per dimension
Blocks	Alternate Current, Enhanced Block Entities (NeoForged)
Structure loading	Structure Layout Optimizer
World	Chunky pregen (starter only), on-demand pregen with a disk cap
Distant Horizons	Off by default ; optional profile

5.3 The measurement protocol

Measured at every phase gate, written into CHANGELOG.md:

1. **Overworld survey** — render distance 12, 60 seconds, record average and 1% low FPS.

2. **Boss arena** — Maledictus arena with a full particle load.
3. **Water stress** — an ocean with Dynamic Waters + Flowing Fluids + Boatload active, tick time logged.
4. **Entity stress** — a Mob Grinding Utils fan farm running, tick time logged.
5. **RAM** — peak heap after 20 minutes of play.

If any number misses, apply the cut order (DESIGN §6.11) before touching render distance. **Render distance is the last lever, not the first.**

5.4 The cut order

When the budget is missed, cut in this order and re-measure after each cut:

```
Currents of Trade → Sable/Sablewaves → Streams Reflowing  
Comforts → Tool Belt → Farmer's Delight → Boatload → Flow  
Monsters & Spellbooks → SWEM
```

The order is deliberate: the optional world-fulishment extras go before anything that changes combat or worldgen feel, and the magic *addon* goes before the magic *base* mod.

5.5 Spell particle cost

Magic is particle-heavy. The first lever is Iron's Spells' own particle settings, then the client's particle setting — never the render-distance budget.

Part 6 — The "don't forget"

register

Everything that is easy to lose track of, in one place. Each row is checked at M9's gate.

#	Thing	Where it's specified	Status check
1	Wizard scroll trades stripped	DESIGN §6.6	Audit every wizard trade table
2	Spartan Weaponry's spears disabled	DESIGN §3	Config check
3	Hopo Better Mineshaft excluded	MODS-INTERACTIONS §2	Not in index.toml
4	Dynamic Waters needs a new world	DESIGN §13	Water stack locked before worldgen
5	Tectonic off by default	D6	Config check
6	No shaders, ever	DESIGN §12	shaderpacks/empty
7	GhostSwap is our custom mod	DESIGN §3	Built and tested in M1
8	The Eldritch Manuscript is consumed	DESIGN §6.6	Verify in M3c
9	Lich's Bargain villager-blood reworked	MODS-INTERACTIONS §3	KubeJS replacement in M3c
10	Rolling does not cancel casts	M1 decision	Verify in M1, re-verify in M3c

11	Attribute Modify leaves irons_spellbooks alone	MODS- INTERACTIONS §3	M5 test
12	Loot beam colours = Reforged tiers	MODS- INTERACTIONS §3	M5 test
13	One music layer only (Boss Music Mod)	MODS- INTERACTIONS §3	M6 test
14	One damage-number layer only (Floating Damage Indicators)	MODS- INTERACTIONS §3	M6 test
15	Nine runes, all findable, none soft-locked	DESIGN §8.3	M7 gate
16	All 12 Annals required for the Cathedral	DESIGN §8.1	M7 gate
17	24 journal pages exist and are obtainable	LORE §10	M7 gate
18	24 Codex entries written	DESIGN §5.5	M7 gate
19	Every signature weapon has an imbued slot	DESIGN §11	M8 gate
20	Ashen material has one source	DESIGN §8.3	M8 gate
21	Act I shippable alone	D2	M7 gate
22	Cut order written into the pack README	DESIGN §6.11	M9 gate
23	Port-risk register re-checked	MODS- INTERACTIONS §7	M9 gate

24	Docs still describe what shipped	all five docs	M9 gate
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Part 7 — Oversight cadence

What you see, and when:

Point	What you get
End of each phase	The gate result table + the CHANGELOG entry + any decision the phase forced
End of M1	A playable combat test — movement and spear feel is your call, not ours
End of M3c	A world seed + coordinates to fly to and judge the Dungeon Layer density
End of M7	Act I playable, chapters 0–5, with the story gates intact
End of M9	Full numbers: FPS, tick time, RAM, mod count, and the completed registers

Your approval points are M1, M3c, M7 and M9. Everything else runs without a checkpoint unless a gate fails.

Phase deliverables and gates are in PHASES.md. Per-mod verification steps are in MODS-INTERACTIONS.md §8. The mod stack itself is DESIGN.md §6.